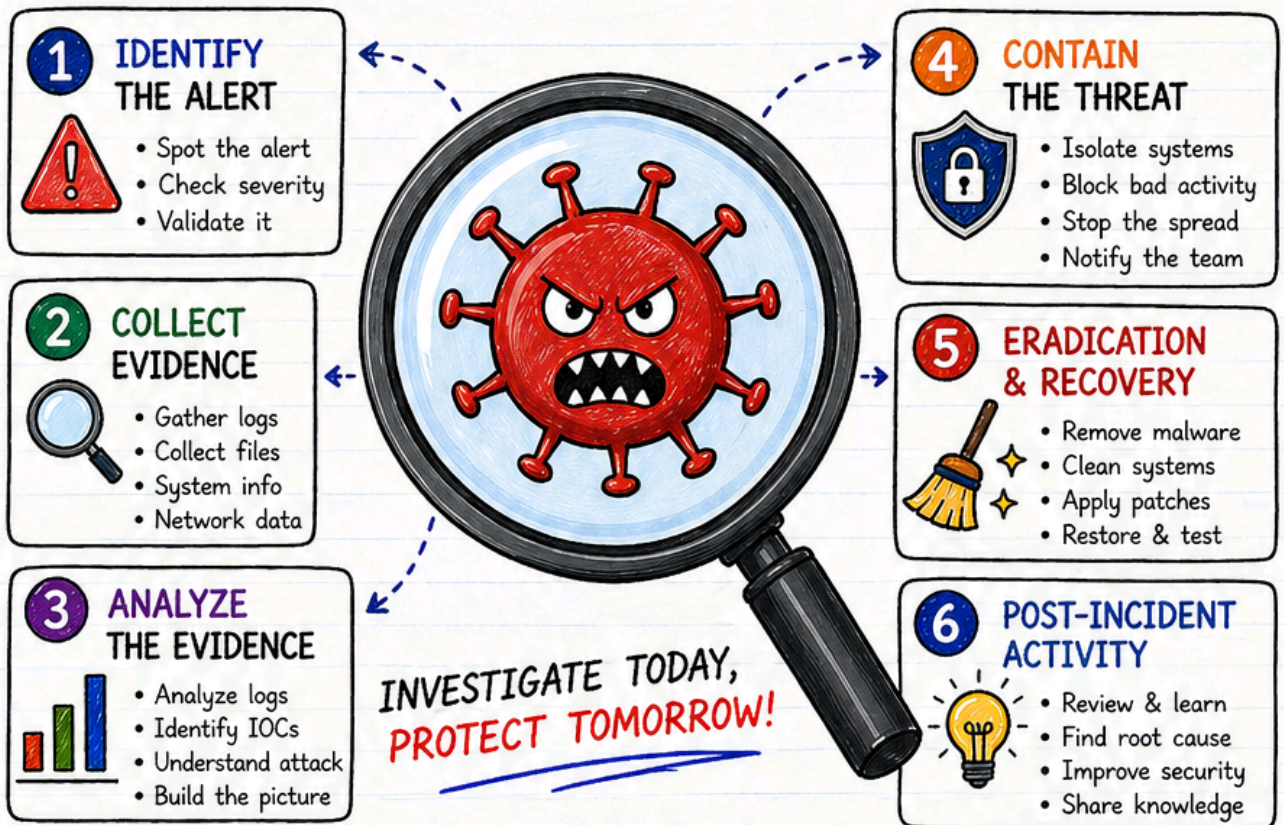




MALWARE INVESTIGATION

✓ STEP-BY-STEP PROCESS | ✓ PRACTICAL EXAMPLES | ✓ REAL-WORLD SCENARIOS



MONITOR
Stay alert
24/7



INVESTIGATE
Dig deep,
find the truth



PROTECT
Strengthen
defenses



SECURE
Secure systems
& data



RESPOND
Act fast,
minimize impact



DETECT
Find threats
early



INVESTIGATE
Dig deep,
find the truth



RESPOND
Act fast,
minimize impact



LEARN
Improve and
stay ahead



TEAMWORK
Stronger together,
safer together

WHAT IS MALWARE?

Malware is short for
"Malicious Software".

It is any bad program or file that is made to harm your computer, steal your data or cause problems.



1 WHY IS MALWARE CREATED?

Hackers (**bad people**) create malware to do bad things.

- To steal your personal information.
- To get money from you.
- To damage your computer or system.
- To take control of your computer.
- To spy on you without your knowledge.
- To attack big companies or websites.



2 WHAT ARE THE MAIN OBJECTIVES OF MALWARE?



Steal Information

Steals passwords, photos, bank details and other private information.



Gain Access

Gets into your computer to control it.



Make Money

Ransomware and other malware ask for money.



Cause Damage

Deletes files, crashes systems or stops them from working.



Spy Secretly

Watches what you do, takes screenshots or records your activities.

3 HOW IS MALWARE DIFFERENT FROM NORMAL SOFTWARE?

Normal Software (Good)



- Made to help us.
- Does useful and safe work.
- We install it with permission.
- It does not harm our system.

VS

Malware (Bad)



- Made to harm us.
- Does bad and harmful work.
- It comes without permission.
- It can damage or steal data.

4 WHAT IS THE DIFFERENCE BETWEEN MALWARE AND EXPLOIT?

Malware



- It is the bad program or file.
- It runs on your system.
- It does harmful things like stealing data or damaging files.



Exploit

- It is a tool or code.
- It finds a weakness (hole) in software.
- It is used to get in and deliver malware.

QUICK QUESTIONS

- 1 What is malware?
- 2 Why do hackers create malware?
- 3 What are the main objectives of malware?
- 4 How is malware different from normal software?
- 5 What is the difference between malware and exploit?



2. TYPES OF MALWARE

Malware comes in many types. Each type works in a different way but the goal is always to harm you.

1 VIRUS



A virus attaches to a file and spreads to other files when you open it. It can delete files or slow down your computer.

2 WORM



A worm is a malware that can spread by itself over networks without any help. It can make your system very slow.

3 TROJAN HORSE



It looks like a safe and useful program but inside it has bad code. It gives hackers access to your computer.

4 RANSOMWARE



It locks your files and shows a message. It asks for money to unlock your files.

5 SPYWARE



It secretly watches what you do on your computer. It can steal your passwords, photos, and other details.

6 ADWARE



It shows many ads on your screen. It is not very dangerous but very annoying.

7 ROOTKIT



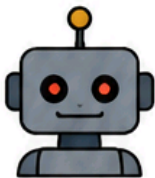
It hides inside your system deeply. It hides other malware and gives full control to hackers.

8 KEYLOGGER



It records every key you type. It can steal your passwords, card numbers and messages.

9 BOT



It turns your computer into a robot. Hackers use it to do bad things without your knowledge.

10 FILELESS MALWARE



It does not use files. It runs in your computer memory and is hard to detect.

QUESTIONS TO THINK

- 1 What is a virus?
- 2 How is a worm different from a virus?
- 3 What does a Trojan Horse do?
- 4 Why is ransomware dangerous?
- 5 Which malware secretly watches you?
- 6 What is the job of a keylogger?
- 7 Can adware harm your computer?
- 8 Why is fileless malware hard to detect?



3. HOW MALWARE ENTERS AND SPREADS

Malware needs a way to get into your computer.
Once it gets in, it can spread and cause damage.

HOW MALWARE ENTERS YOUR COMPUTER



1 Email

Through bad links or infected attachments in emails.



2 Internet Downloads

From unsafe websites or fake software downloads.



3 USB Devices

From infected pen drives, external hard drives or other devices.



4 Weak Security

When your antivirus is off or your system is not updated.



5 Instant Messages

Through bad links or files shared in chats or messages.



6 Fake Software

From cracked software or keygens that contain malware.

HOW MALWARE SPREADS AFTER ENTERING



1 Through Network

It moves from one computer to another in the same network.



2 Through Files

It attaches itself to files and spreads when files are opened.



3 Through Email

It sends copies of itself to other people's emails.



4 Through Websites

It uses unsafe websites to download more malware.



5 Through System Vulnerabilities

It uses weaknesses (holes) in software to spread.

THINGS TO REMEMBER



- ✓ Be careful with emails and links.
- ✓ Download only from trusted websites.
- ✓ Do not use unknown USB devices.
- ✓ Keep your antivirus on.
- ✓ Always keep your system updated.

QUESTIONS TO THINK

- 1 How does malware enter your computer?
- 2 Which are the most common ways malware spreads?
- 3 Why is it important to keep our software updated?
- 4 What should you do if you get a suspicious email?



4. HOW MALWARE DAMAGES YOU

Malware is made to harm you.

It can steal your data, spoil your files, slow down your computer and cause many other problems.



MAIN WAYS MALWARE DAMAGES YOUR COMPUTER AND YOU

1 Steals Information



It steals your passwords, photos, bank details and other personal data.

2 Locks Your Files



It locks your files and asks for money to unlock them (ransomware).

3 Slows Down Your Computer



It uses many system resources and makes your computer very slow.

4 Deletes or Damages Files



It deletes, changes or damages your important files.

5 Takes Control of Your System



It gives hackers full control to do bad things on your computer.

6 Spies on You



It watches what you do, takes screenshots or records your activities.

7 Shows Annoying Ads



It shows too many ads and pop-ups on your screen.

8 Breaks Internet Connection



It changes your internet settings or blocks websites.

9 Causes Money Loss



It can steal money from your bank or make you pay ransom.

10 Makes Your System Unstable



It causes errors, crashes or even stops your computer from starting.

OTHER BIG PROBLEMS

- You can lose important work.
- Your private life is not safe.
- Your friends or contacts can be harmed.
- Your devices can be used for attacks on others.



EXAMPLES IN REAL LIFE

- A company lost all files and had to pay lots of money to get them back.
- Someone's bank details were stolen and money was taken.
- A hospital system was attacked and doctors could not use computers.
- A student lost all project files due to ransomware.



QUESTIONS TO THINK

- 1 What can malware steal from you?
- 2 How can malware affect your files?
- 3 Why does malware slow down your computer?
- 4 What is ransomware?
- 5 What other big problems can malware cause?

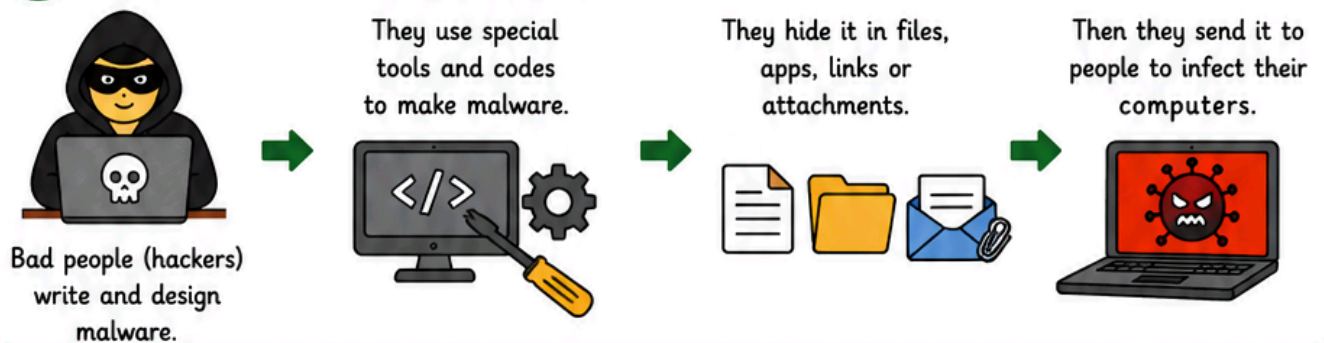


5. HOW MALWARE IS CREATED, DESTROYED AND AI ROLE

Malware is made by bad people to harm computers. But good people (security experts) work to destroy it and keep us safe.



1 HOW IS MALWARE CREATED?



2 HOW IS MALWARE DESTROYED?



3 HOW AI HELPS ATTACKERS AND DEFENDERS?

AI helps Attackers (Bad Use)

- AI helps them write malware faster.
- AI helps them hide malware so it is hard to detect.
- AI helps them find weak spots in systems automatically.
- AI helps them send fake emails that look real.



AI helps Defenders (Good Use)

- AI finds malware quickly.
- AI can spot strange activities in real time.
- AI can block attacks before they cause damage.
- AI helps in automatic updates and threat alerts.



QUESTIONS TO THINK

- 1 Who creates malware and why?
- 2 How do hackers hide and spread malware?
- 3 How do security experts destroy malware?
- 4 What tools do they use to remove malware?
- 5 How does AI help attackers?
- 6 How does AI help defenders?
- 7 Which side (good or bad) do you think AI helps more? Why?



MALWARE INVESTIGATION

STEPS (SOC LEVEL)

When we think a computer has malware, we follow steps to find out:
What happened? How? And what to do next?



STEP 1 IDENTIFY AND CONFIRM THE ALERT

WHAT WE DO?

- We first get an alert from antivirus, EDR, firewall, email gateway or SIEM (security tool).
- We read the alert carefully.
- We check what is the problem, where it happened and when.



EXAMPLE ALERTS

- Antivirus detected a virus.
- Suspicious file was executed.
- Malicious activity was blocked.
- Ransomware behavior detected.



Antivirus Alert



EDR Alert



Firewall Alert



Email Alert

KEY POINTS



- Not every alert is a real attack.
- We must verify before taking action.
- This step helps us save time and work on real threats.

QUESTIONS TO THINK

- 1 What is the first step in malware investigation?
- 2 From where do we get alerts?
- 3 Why is it important to read the alert carefully?



Remember:
Good investigation starts with good information.

STEP 2

COLLECT AND GATHER EVIDENCE

After we get the alert, we collect all useful information to understand the problem better.
Good evidence = good investigation!



WHAT WE COLLECT?

LOGS



We collect logs from firewalls, servers, EDR, applications, and other tools.

FILES



We find and collect the suspicious file (malicious file) if any.

PROCESS INFO



We check running processes to see if anything suspicious is running.

NETWORK INFO



We collect details about network connections, IP addresses and domains.

SYSTEM INFO



We gather system details like OS, username, time, installed software, etc.

HOW WE COLLECT?



From security tools like EDR, SIEM, Firewall.



Using commands on the system (like `ps`, `netstat`, `ipconfig`, etc.).



By checking files, folders and registry entries.



From network captures and threat intelligence.



From backups, databases and other sources.

KEY POINTS



- Collect as much information as possible.
- Do not change or delete anything.
- Good evidence helps us find the real cause.
- Always record where the data came from.



QUESTIONS TO THINK

- 1 What is the main goal of this step?
- 2 What are some things we collect as evidence?
- 3 From which tools can we collect logs?
- 4 Why is it important not to delete or change anything?
- 5 Which commands can we use to collect information?



Remember:

More information
= clearer picture
= better decision!

STEP 3

ANALYZE THE EVIDENCE



Now we study the evidence to understand what happened, how it happened and what kind of malware is involved.

WHAT WE ANALYZE?

FILE ANALYSIS



We check the file type, size, hash, behavior and test it in a safe environment.

LOG ANALYSIS



We read logs to see what happened before, during and after the attack.

NETWORK ANALYSIS



We look at network traffic, connections, IP addresses, domains and ports.

BEHAVIOR ANALYSIS



We see how the malware behaves on the system (what it does and changes).

TIMELINE ANALYSIS



We create a timeline to understand the order of events clearly.

HOW WE ANALYZE?



Open the evidence in a safe tool or sandbox.



Check important details and look for suspicious items.



Use tools and commands to dig deeper (like strings, tasklist, netstat, etc.).



Find patterns, connections and indicators of malicious activity.



Write down what we found and what it means.

KEY POINTS



- The goal is to understand the full picture of the attack.
- Look for "Indicators of Compromise" (IOCs) like IPs, domains, hashes, file names.
- Always be careful and analyze in a safe environment.
- Good analysis helps us find the root cause and stop future attacks.

QUESTIONS TO THINK

- 1 What is the goal of this step?
- 2 What are some things we analyze?
- 3 Why do we use a sandbox or safe environment?
- 4 What are Indicators of Compromise (IOCs)?
- 5 How does timeline analysis help us?



Remember:

Good analysis
= correct understanding
= correct action!

STEP 4

CONTAIN THE THREAT (STOP THE DAMAGE)

After understanding the problem, we take action to stop the malware from spreading and causing more harm.



WHAT WE DO?

ISOLATE AFFECTED SYSTEMS



We disconnect the infected system from the network to stop the spread.

BLOCK MALICIOUS ACTIVITY



We block bad IPs, domains, files and connections using firewall, EDR, or other tools.

END MALICIOUS PROCESSES



We stop harmful processes and services running on the system.

QUARANTINE MALICIOUS FILES



We move dangerous files to quarantine so they cannot cause more harm.

NOTIFY THE RIGHT PEOPLE

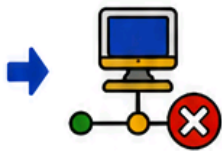


We inform the team, manager or other teams as per the company process.

HOW WE CONTAIN?



Confirm the malware infection.



Isolate the system from the network.



Block malicious IPs, domains and connections.



Stop malicious processes and services.



Quarantine or remove malicious files.



Inform the right people.

KEY POINTS



- Containment helps stop the malware from spreading.
- Quick action can save important data and systems.
- Follow your company's incident response process.
- Always document what actions you took.
- Do not delete evidence before taking backup.



QUESTIONS TO THINK

- 1 What is the main goal of this step?
- 2 How do we stop malware from spreading?
- 3 What tools can help us block malicious activity?
- 4 Why do we isolate the infected system?
- 5 Who should we inform during this step?



Remember:

Containment today
= less damage
tomorrow!

STEP 5

ERADICATION & RECOVERY (CLEAN AND RESTORE)

Now we remove the malware completely from the system and bring everything back to normal safely.



WHAT WE DO?

REMOVE MALWARE



We remove the malware files, bad programs and unwanted changes from the system.

CLEAN SYSTEM



We clean temp files, registry entries, scheduled tasks and startup items.

PATCH & UPDATE



We install missing updates and patches to close security holes.

RESTORE DATA



We restore files and settings from clean backups if anything was lost.

VERIFY & TEST



We check the system carefully to make sure everything is working fine.

HOW WE DO IT?



Confirm the infection is gone.



Remove malware and related elements.



Clean system and remove unwanted entries.



Install updates and security patches.



Restore data from clean backups (if needed).



Test and monitor to ensure system is normal.

KEY POINTS



- Eradication removes the root cause of the malware.
- Recovery brings the system back to normal safely.
- Always use trusted tools and clean backups.
- Do not skip updates and patches.
- Verify and test before giving access back to users.



QUESTIONS TO THINK

- 1 What is the main goal of this step?
- 2 What do we remove during eradication?
- 3 Why are updates and patches important?
- 4 When do we restore data from backups?
- 5 Why do we verify and test at the end?



Remember:
Clean today,
Secure tomorrow,
Safe always!

STEP 6

POST-INCIDENT ACTIVITY (LEARN AND IMPROVE)

After the incident is over, we review everything to learn from it and make our security better so that we can stop similar attacks in the future.



WHAT WE DO?

REVIEW THE INCIDENT



We review what happened, how it happened and what kind of malware was used.

FIND ROOT CAUSE



We find the main reason how the malware got in and what mistakes allowed it.

IMPROVE SECURITY



We update rules, tools, settings and processes to make our systems stronger.

UPDATE DOCUMENTATION



We update all records, reports, playbooks and lessons learned.

SHARE AND TRAIN



We share knowledge with the team and do training to be ready for the next time.

HOW WE DO IT?



Review all logs, evidence and actions taken.



Find the root cause of the attack.



Improve tools, rules and security controls.



Update documents, reports and playbooks.



Train the team and share what we learned.

KEY POINTS



- Post-incident activity helps us learn and improve.
- It reduces the chance of the same attack happening again.
- Good documentation helps the team and future investigations.
- Continuous improvement makes our security strong.



QUESTIONS TO THINK

- 1 What is the main goal of this step?
- 2 Why is it important to find the root cause?
- 3 What are some things we improve after an incident?
- 4 Why should we update documentation and playbooks?
- 5 How does sharing and training help the team?



Remember:
Learn today,
Improve today,
Stronger tomorrow!
Stay one step ahead!

MALWARE INVESTIGATION

OVERALL SUMMARY

Malware investigation is the process of finding, analyzing and removing malware from a system and learning from it to make our security stronger.



THE 6 KEY STEPS AT A GLANCE

STEP	STEP NAME	MAIN GOAL	KEY ACTIONS (IN SHORT)	OUTCOME
1 	IDENTIFY THE ALERT	Know that something is wrong.	- Monitor alerts from tools - Check severity & details - Validate if it is real	We confirm a possible security incident.
2 	COLLECT EVIDENCE	Collect all useful information.	- Collect logs, files, process, network & system info - Use tools and commands - Do not delete or change data	We have good evidence to investigate.
3 	ANALYZE THE EVIDENCE	Understand what happened and how.	- Analyze files, logs, network and behavior - Find IOCs and patterns - Build the full picture	We know what happened and the malware involved.
4 	CONTAIN THE THREAT	Stop the malware from causing more damage.	- Isolate infected systems - Block bad activity (IPs, domains) - Stop malicious processes - Notify the right people	The threat is stopped and contained.
5 	ERADICATION & RECOVERY	Remove the malware and bring systems back to normal.	- Remove malware & bad programs - Clean system & apply patches - Restore data (if needed) - Verify and test the system	System is clean, restored and working safely.
6 	POST-INCIDENT ACTIVITY	Learn from the incident and improve security for the future.	- Review what happened - Find root cause and mistakes - Improve rules, tools & processes - Share knowledge & train team	We become stronger and better prepared.

IMPORTANT REMINDERS



Always preserve evidence.



Act quickly to contain threats.



Follow your company's process and guidelines.



Keep systems, tools and data up to date.



Teamwork and communication are key.

THE BIG PICTURE



Our goal is simple: **Detect** early, **Investigate** properly, **Respond** quickly and **Learn** continuously.
This protects our systems, data and organization!



MALWARE INVESTIGATION - YOU'VE GOT THE SKILLS!

★ Great Work! You have completed your notes on **MALWARE INVESTIGATION**. ★



You now know how to Detect, Investigate, Respond and Learn.
Keep practicing, stay curious,
and you will become an amazing SOC Analyst! ❤️

MALWARE INVESTIGATION AT A GLANCE

- 1 Identify the alert
- 2 Collect evidence
- 3 Analyze the evidence
- 4 Contain the threat
- 5 Eradication & recovery
- 6 Post-incident activity (Learn & improve)



KEY TAKEAWAYS

- ★ Early detection saves time and data.
- ★ Good evidence leads to good answers.
- ★ Analysis turns data into understanding.
- ★ Quick containment stops more damage.
- ★ Clean systems + updates = safe systems.
- ★ Learning from incidents makes us stronger.



INVESTIGATION FLOW - THE 6 STEPS



Identify the alert



Collect evidence



Analyze the evidence



Contain the threat



Eradication & recovery



Post-incident (Learn & improve)

TOP TIPS FOR ANALYSTS

- ✓ Be curious and ask questions.
- ✓ Always verify before you act.
- ✓ Document everything you do.
- ✓ Use the right tools and techniques.
- ✓ Stay calm and follow the process.



TOOLS YOU WILL USE (EXAMPLES)

- ★ SIEM (Splunk, Wazuh, Elastic)
- ★ EDR (Microsoft Defender, CrowdStrike)
- ★ Network Tools (Wireshark, Zeek)
- ★ Forensics Tools (Autopsy, FTK)
- ★ OS Tools (PowerShell, Cmd, Linux CLI)



CHALLENGES YOU WILL FACE (AND OVERCOME!)

- ✓ Complex attacks → Break them down step by step.
- ✓ Large data & alerts → Focus on what matters.
- ✓ No clear answers → Use logic, evidence and experience.
- ✓ Time pressure → Stay calm, follow the process.
- ✓ Keep learning → Every incident makes you better!



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the stronger you become.
The stronger you become,
the more you can protect."

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- Keep Practicing. ✓
- Keep Defending. ✓



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